

SMART REAL-TIME TAXI FARE COMPARISON AND MACHINE LEARNING  
BASED FARE INTELLIGENCE SYSTEM

COMPREHENSIVE PROJECT PROBLEM STATEMENT & SPECIFICATION

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| Candidate Name:   | SIDDHARTH MV                                          | Degree / Program: | Master of Computer Applications (MCA)           |
| Project Platform: | RideCompare                                           | Academic Year:    | 2026 – 2027                                     |
| Domain:           | Urban Mobility / Applied ML & Distributed Web Systems | Document Scope:   | Formal Problem Statement & Technical Objectives |

1. Background & Urban Mobility Landscape

Over the past decade, urban passenger transportation across metropolitan economies has undergone a structural transition from traditional street-hail taxis toward on-demand ride-hailing aggregator platforms. Commercial ride-hailing networks—prominently Uber, Ola Cabs, and Rapido in India—alongside indigenous metered auto-rickshaws and state-regulated public taxi bodies, constitute the primary modal choice for millions of daily commuters. These platforms operate on dynamic pricing algorithms that continuously modulate journey fares in response to micro-fluctuations in passenger demand, driver supply, traffic congestion density, weather events, and specific diurnal transit windows. While dynamic pricing serves an economic purpose by clearing supply-demand imbalances in real time, it creates severe market opacity from the perspective of the individual consumer. Ride fares fluctuate unpredictably, often varying by 40% to 150% across competing platforms for identical journey corridors at the exact same minute. Consequently, urban commuters are left without transparent tariff benchmarks or unified comparison utilities.

2. The Real-World Commuter Problem

Under the current urban transit paradigm, an individual commuter attempting to book a ride must undertake an inefficient, iterative manual process: (1) launch the first ride-hailing app (e.g., Uber) and enter pickup location; (2) enter destination and wait for geocoding and fare quotation; (3) review vehicle options and note fares and ETAs; (4) switch to a second provider (e.g., Ola) and repeat the entire input process from scratch; (5) switch to a third application (e.g., Rapido) for bike or auto options; and (6) mentally compare the fragmented results, balance price vs. ETA trade-offs, and make a decision under time pressure.

3. Formal Problem Statement

Problem Statement:

"To design, develop, evaluate, and deploy a centralized, production-grade web application—entitled '**Smart Real-Time Taxi Fare Comparison and Machine Learning Based Fare Intelligence System' (RideCompare)**—that concurrently aggregates multi-provider taxi tariffs, determines road routing kinetics, normalizes disparate vehicle categories, and leverages in-process Machine Learning models (K-Means clustering, Gradient Boosting regression, and Isolation Forest anomaly detection) to provide transparent fair tariff baselines, surge explanations, and multi-factor ride rankings in real time."

## 4. Deficiencies and Pain Points of Existing Systems

- In the existing commercial ecosystem, service providers operate as isolated walled gardens. Analysis of consumer workflows highlights six acute deficiencies:
- Fragmented Information:** Tariff details are isolated within proprietary mobile applications, preventing unified market visibility and forcing commuters into tedious app-switching.
  - Rapidly Expiring Prices:** Dynamic quotes typically possess short expiration windows (60 seconds). By the time a user finishes comparing three apps, earlier quotes have already expired or repriced.
  - Surge Pricing Opacity:** Aggregator apps mask algorithmic markups behind simple surge multipliers without explanatory diagnostics. Commuters cannot discern whether a high quote reflects genuine congestion or opportunistic price gouging.
  - Disparate Vehicle Categories:** Comparing a bike taxi from one provider against an auto-rickshaw or compact sedan from another requires complex mental normalization of speed, comfort, and weather risk.
  - Lack of Historical Context:** Existing applications present point-in-time quotes without context. Commuters have no way of knowing whether a quoted fare is historically normal, elevated, or an anomalous pricing glitch.
  - Omission of Public & Metered Transit:** Commercial aggregators deliberately omit government-regulated metered taxis and autos that maintain fixed statutory rates with zero surge pricing.

## 5. Specific, Measurable Project Objectives

- To address each facet of the problem statement, the project establishes eight rigorous, measurable technical objectives:
- 1. Data Collection & Preprocessing:** Assemble and preprocess a structured historical dataset of 12,000 transit records across Indian metropolitan corridors, handling physical bounds, cyclic diurnal features, and unit economic derivations.
  - 2. Concurrent Quote Orchestration:** Construct an asynchronous backend architecture capable of querying multi-provider rate cards (Uber Go, Uber Premier, Ola Mini, Ola Prime, Rapido Bike, Rapido Auto, Local Taxi) within a 15-second freshness window.
  - 3. Geospatial Road Kinematics:** Integrate open geospatial services (OpenStreetMap Nominatim for geocoding and Project-OSRM for turn-by-turn road geometry) to obtain accurate route distances and driving durations.
  - 4. Unsupervised Pricing Regime Discovery:** Train an unsupervised K-Means clustering model to classify rides into empirical pricing regimes ('Peak Hour Surge', 'Standard City Transit', 'Long-Distance Transit') with silhouette validation.
  - 5. Supervised Fair Baseline Estimation:** Train, evaluate, and tune supervised regression models (Random Forest vs Gradient Tree Boosting) to predict an expected fair tariff baseline with high statistical fidelity ( $R^2 > 0.95$ ).
  - 6. Multivariate Anomaly Detection:** Deploy an Isolation Forest detector to flag abnormal pricing spikes and fare glitches with natural-language diagnostic feedback.
  - 7. Composite Smart Scoring:** Formulate a multi-criteria utility score balancing fare (40%), pickup ETA (30%), prediction confidence (15%), and provider reliability (15%).
  - 8. Responsive Web Interface:** Develop a high-performance React 19 / TypeScript single-page application featuring interactive Leaflet route mapping, corridor price volatility sparklines, and direct app deep linking.

6. Project Scope and Operational Boundaries

- **Target Users:** Daily office commuters, university students, budget-conscious travelers, airport passengers, and urban transport analysts.
- **Supported Locations:** Primary coverage encompasses Indian metropolitan transit hubs: Bangalore (default), Delhi NCR, and Mumbai, with arbitrary coordinate-based routing supported up to 300 km.
- **Supported Services:** Uber Go, Uber Premier, Ola Mini, Ola Prime, Rapido Bike, Rapido Auto, and Local Metered Taxis.
- **Data Scope:** 12,000 empirical historical records encompassing 22 features, reflecting multi-tier vehicle categories, traffic levels, weather conditions, and diurnal commute peaks.
- **Machine Learning Scope:** In-process inference running synchronously with quote retrieval, achieving sub-millisecond execution times without external microservice overhead.
- **Operational Boundary:** As commercial aggregators do not provide open, public API endpoints for real-time third-party booking without proprietary enterprise contracts, live provider quotes are computed using calibrated statutory rate cards matching published tariff regulations, integrated with universal app deep-linking.

7. Proposed System Architecture & Core Advantages

RideCompare solves the problem via a decoupled three-tier system: React 19 UI with Leaflet, FastAPI backend gateway with Project-OSRM, and in-process Scikit-Learn ML intelligence pipelines.

- **Single Search Query:** Origin and destination are specified once; geocoding and road routing are resolved automatically.
- **Sub-Second Aggregation:** Concurrent asynchronous orchestration delivers full multi-provider comparison in under 600 ms (90% search latency reduction vs. manual search).
- **Quantified Surge Transparency:** The ML baseline displays the exact numerical surge premium (e.g., '+Rs. 54.20 (+28% surge)') over the fair expected fare.
- **Multi-Modal Coverage:** Bikes, auto-rickshaws, compact sedans, premium cabs, and local metered taxis are compared side-by-side.
- **Multi-Factor Smart Utility:** Rides are ranked not only by price, but by a balanced utility formula incorporating ETA, ML prediction confidence, and reliability.
- **One-Tap Direct Transition:** Universal deep links allow instant handover to native apps with coordinates pre-populated.

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